Cognex Corporation

CGNX | Nasdaq | 2026-10-07

Q4 2027 model ceiling: USD 43.53

Analytical Base 43/60 = 72 | Watchlist

20x quarter-plus-next-three fully expensed GAAP operating profit plus fixed opening excess liquidity, less matched transaction cash in closing states, divided by state-specific end-window diluted shares; common-equity values are probability weighted after calculation. No future retained cash credit.

Coverage initiated at the owner's request. No holding was established by this research. Existing portfolio Base overrides remain protected.

All operating values are in USD m and shares in millions unless stated otherwise. Forecasts are analytical assumptions; issuer guidance and actual results are identified separately.

Writing-date snapshot: USD 61.21 per share (2026-10-07 close); 168.224 million common shares (2026-08-02); estimated equity capitalization USD 10,297.0 million. Exact filing-dated issued shares, not an assertion of an October 7 share count. This dated common-share snapshot is separate from forecast diluted shares.

Decision in the model

Active horizon	Earnings window	F4Q OP	Multiple	Diluted shares	Ceiling
Q4 2027	Q4 2027 through Q3 2028	374.6	20x	174.305	43.53

The ceiling is a model valuation, not a guaranteed trading price or a literal net-income P/E. Operating branches are weighted once. Where debt, preferred claims, conversion or equity floors make value nonlinear, the model values each coherent operating and funding state first, then weights those common-share outcomes.

Economic scope and matching claims

Cognex consolidated core plus RealSense only within closing states. Financial debt is zero; lease rent and ordinary operating claims remain in fully expensed OP. Fixed opening USD 555.013 million excess-liquidity credit consumes USD 510 million acquisition/transaction cash in closing outcomes.. 20x quarter-plus-next-three fully expensed GAAP operating profit plus fixed opening excess liquidity, less matched transaction cash in closing states, divided by state-specific end-window diluted shares; common-equity values are probability weighted after calculation. No future retained cash credit.

Financial claims and diluted shares follow the company-specific schedule below. Debt, preferred and minority claims are counted once. All financial model tables use USD m unless a different unit is stated.

Thesis

Cognex sells the ability to identify, measure and inspect an object reliably enough to change a production decision. The customer pays for a vision system, software and an application that can reject the wrong part, read a damaged code, locate an object or verify an assembly. The relevant economic unit is an accepted inspection installation or qualified OEM design, not a robot, a semiconductor wafer or a data-center rack in its entirety.

The thesis has two distinct parts. The established precision-vision business is recovering across logistics, packaging, electronics and semiconductor equipment while a simpler cost structure improves profitability. The proposed RealSense acquisition adds robotic depth perception, but also substantial retention expense, dilution and acquisition amortization. This analysis values fully expensed operating profit and each acquisition outcome separately. It does not apply the seller's long-term adjusted margin to the acquired revenue immediately.

The analytical Base is **72/100, Watchlist**. There is useful engineering differentiation and verified operating recovery, but short order visibility, capable competitors and uncertain acquisition economics limit conviction. The probability-weighted Q4 2027 ceiling is **USD 43.53 per share**, using 20x the Q4 2027-Q3 2028 fully expensed operating-profit window and the cash and diluted shares belonging to each outcome. It is a model valuation, not management guidance or an imported analyst target.

The product and engineering advantage

A camera does not solve an inspection problem by itself. Illumination must expose the relevant surface feature; optics must deliver sufficient contrast and focus; exposure must freeze motion; and software must distinguish acceptable variation from a defect within the available cycle time. Better inference cannot reliably reconstruct a defect that never appears in the optical signal. A qualified solution therefore combines image formation, inference, tools for the application engineer and a maintained production workflow.

Cognex's established products cover DataMan barcode reading, In-Sight measurement and inspection, VisionPro software, and associated optics and lighting, including Moritex. In-Sight 3900 integrates embedded processing in a compact form factor; the modular 6900 separates more of the camera, lens and lighting choices. The Q1 presentation identifies Qualcomm Dragonwing in the 3900 and NVIDIA Jetson in the 6900. These are complementary compute platforms inside a Cognex application, not evidence that Cognex owns the underlying accelerator technology. The slide's faster-performance claim lacks enough workload conditions to treat it as a universal benchmark.



Embedded AI systems have distinct form factors. The 3900 uses Qualcomm Dragonwing and the 6900 uses NVIDIA Jetson; the 4x statement is issuer marketing under unspecified comparative conditions.

Original Cognex Q1 2026 presentation, page 4. Different form factors change integration choices. The issuer's relative speed claim is not a guarantee for every inspection workload, resolution or accuracy target.

OneVision moves more training, collaboration and deployment management into a common software workflow while inference remains at the edge. Management described hundreds of users and general availability during the Q2 discussion. That can reduce the time required to create and maintain applications, enlarge the set of economically inspectable defects and strengthen customer familiarity with Cognex tools. It does not establish a separately disclosed recurring-software revenue stream. This model retains the hardware-and-application economics rather than importing a subscription multiple.

The strongest barrier is the cost of qualifying a replacement into a live process: false acceptance can ship a defective product, false rejection wastes yield, and downtime disrupts the line. That barrier is strongest in demanding, validated OEM designs and weaker for generic code reading or an easily reproduced demonstration. KEYENCE markets capable AI and rule-based vision systems; customers can also develop internal applications using open software and commodity compute. Better tools increase the addressable market and lower some entry barriers at the same time. The investment case therefore requires paid deployments and retained gross profit, not the assertion that AI eliminates competition.

Who pays, and where the incremental wallet sits

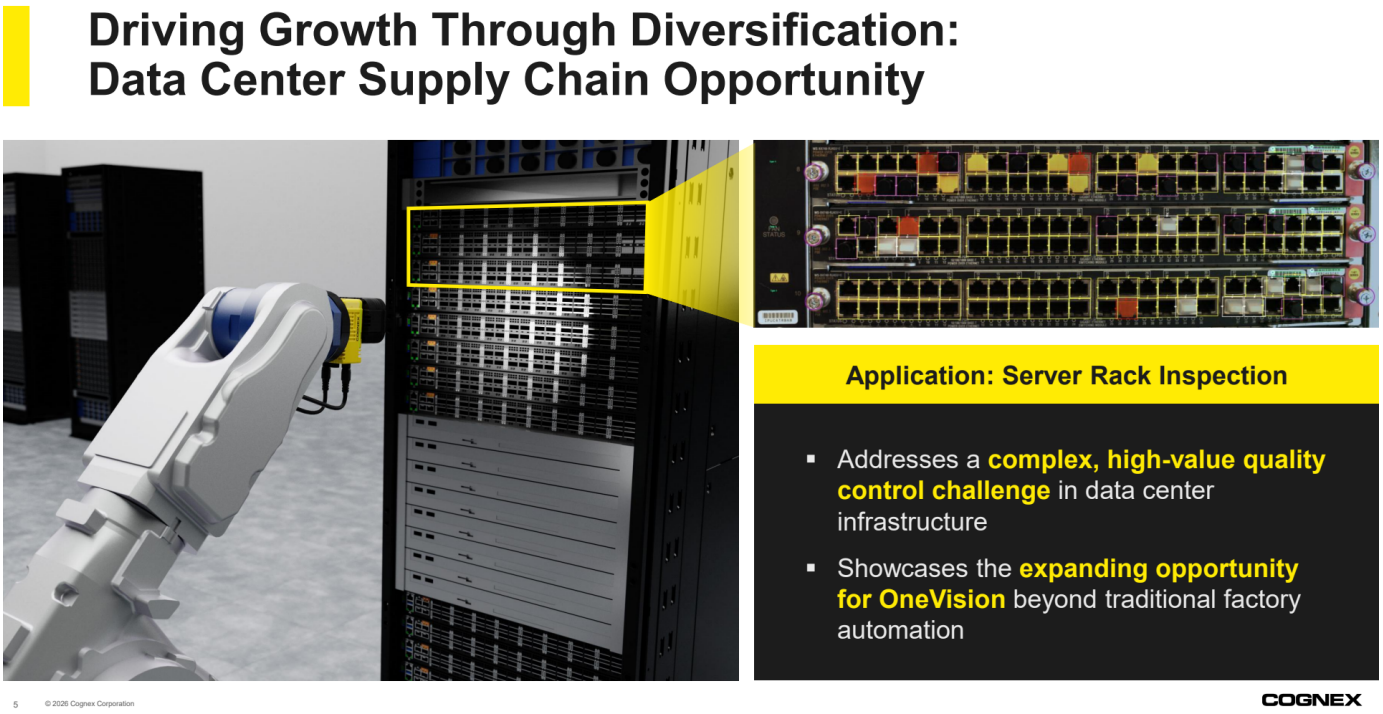
Logistics remains predominantly a barcode-reading business. SLX extends the inspection wallet toward difficult package flows, side-by-side parcels, damage recognition and optical character reading. Management described a price premium for more valuable capabilities. That premium is credible as a bounded branch because missed shipments and labor have measurable costs. It is not permission to apply an inspection-system premium to every parcel, conveyor or historical reader.

Packaging pays for identification, label verification and defect inspection. Consumer electronics pays for exacting assembly and quality-control tasks, but product-launch cycles can make orders uneven. Automotive inspection benefits from automation while remaining exposed to weak vehicle investment. Semiconductor OEM customers qualify machine-vision components and optics into their equipment; Cognex earns its own installed content, not a share of the fab's wafer economics. These product uses explain the revenue vectors below, although Cognex itself reports a single operating segment.

The company reported more than 4,500 new customers in the first half of 2026 and approximately 9,000 during 2025. Its acquisition release describes over 30,000 customers. These are distribution and adoption indicators, not equal-sized revenue cohorts. One unnamed customer represented 15% of 2025 revenue, versus 10% in 2024. Its identity is not established by the filing and is not inferred here. Customer breadth does not eliminate concentrated ordering risk.

Data-center quality control

Management described data-center exposure as a low-single-digit percentage of revenue growing more than 30%. Most existing work inspects components such as boards, connectors and metal parts; server-rack assembly inspection is an emerging application. A GPU shipment, rack power rating or data-center capital budget cannot be converted directly into Cognex revenue without the number of paid inspection stations, the content per station, acceptance and the customer's procurement choice.



The invoice is a vision system and application, not the rack. The pictured deployment is server-rack inspection; most current data-center revenue remains component inspection.

Original Cognex Q2 2026 presentation, page 5. This is an inspection application, not evidence that Cognex supplies the complete rack or captures its capital value.

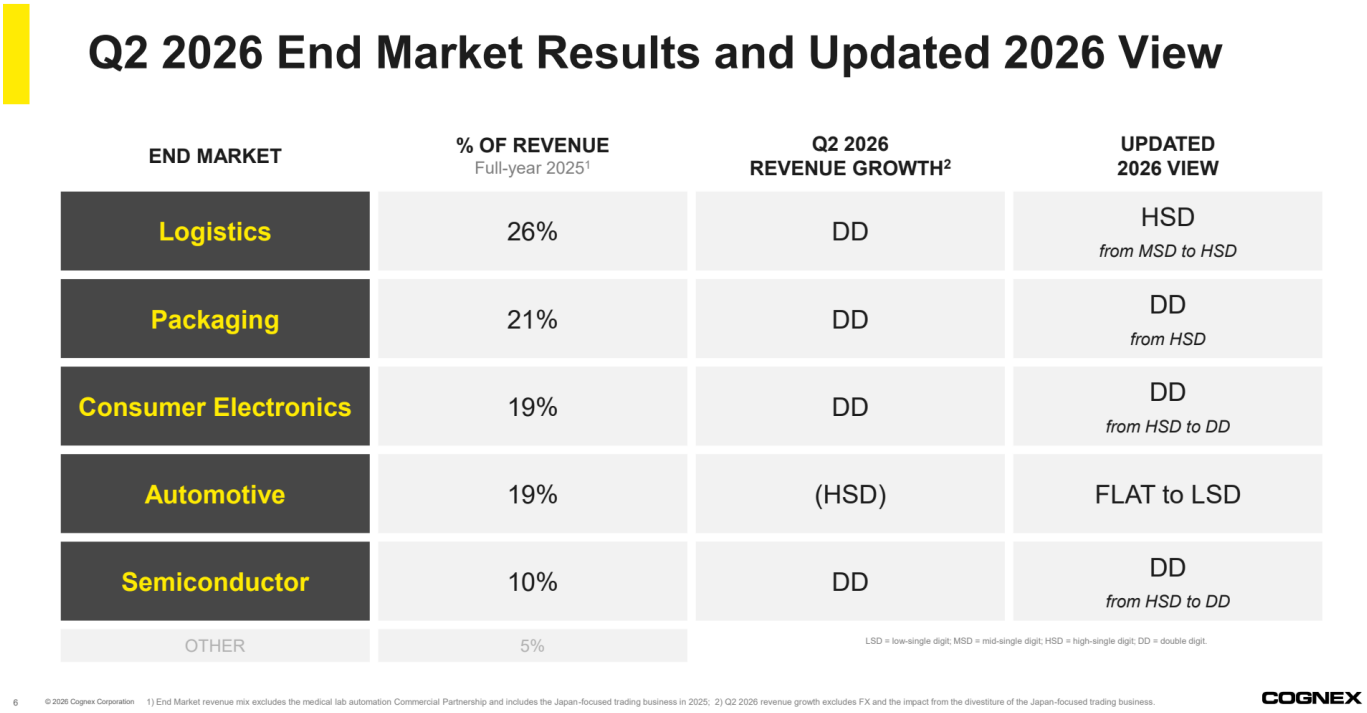
The reference assigns USD 30 million of 2026 core revenue to data-center supply-chain inspection, approximately 2.6% of total modeled core sales. It is carved out of the existing company total rather than added on top of electronics and semiconductor revenue. Growth of 30% in 2027, 25% in 2028 and progressively lower rates

thereafter reflects adoption without assuming every AI factory uses Cognex. No additional AI demand shock is layered onto the same invoices.

Reported performance and reference calibration

The quarter ended July 5, 2026 produced revenue of USD 291.263 million, gross profit of USD 205.773 million, GAAP operating profit of USD 85.517 million and net income of USD 72.756 million. First-half revenue was USD 559.700 million and operating profit USD 145.390 million. Q2 adjusted EBITDA was USD 93.659 million; first-half adjusted EBITDA was USD 165.896 million. Ordinary depreciation, purchase amortization and other adjustments explain the difference. Stock compensation remains in the company's adjusted figures and must not be subtracted a second time from those figures.

Q2 management described double-digit growth in logistics, packaging, electronics and semiconductor applications, with weaker automotive revenue. Geographic comparisons also reflect customer procurement-location changes, so unusually strong China growth is not automatically an equivalent end-demand acceleration. Roughly two-thirds of annual revenue is outside the United States. Currency, tariffs, customer capital budgets and local competitors therefore remain economically relevant.



Q2 2026 End Market Results and Updated 2026 View

END MARKET	% OF REVENUE Full-year 2025[1]	Q2 2026 REVENUE GROWTH[2]	UPDATED 2026 VIEW
Logistics	26%	DD	HSD *from MSD to HSD*
Packaging	21%	DD	DD *from HSD*
Consumer Electronics	19%	DD	DD *from HSD to DD*
Automotive	19%	(HSD)	FLAT to LSD
Semiconductor	10%	DD	DD *from HSD to DD*
OTHER	5%		

LSD = low-single digit; MSD = mid-single digit; HSD = high-single digit; DD = double digit.

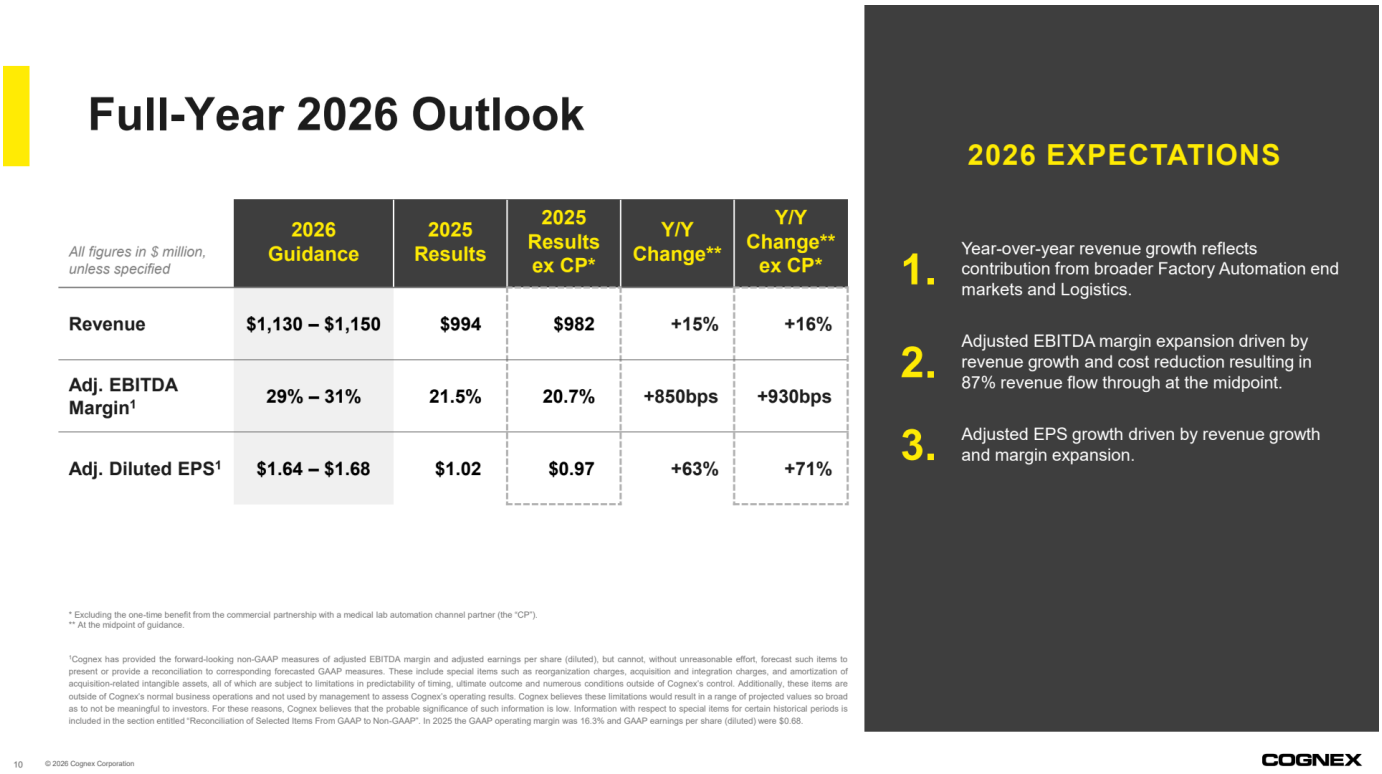
6 © 2026 Cognex Corporation 1) End Market revenue mix excludes the medical lab automation Commercial Partnership and includes the Japan-focused trading business in 2025; 2) Q2 2026 revenue growth excludes FX and the impact from the divestiture of the Japan-focused trading business. COGNEX

2025 end-market shares exclude the commercial partnership and include the divested Japan trading activity. They are not 2026 model segments.

Original Cognex Q2 2026 presentation, page 6. These are 2025 end-market percentages with specified exclusions and the issuer's 2026 outlook. They are not reported 2026 segment accounts.

August guidance was USD 300-320 million Q3 revenue with a 32-35% adjusted EBITDA margin, and USD 1.13-1.15 billion full-year revenue with a 29-31% adjusted EBITDA margin. It predates the RealSense announcement. The core reference uses the USD 1.14 billion midpoint and USD 342 million full-year adjusted EBITDA, exactly 30%. With first-half reported results and USD 310 million Q3 revenue, Q4 core revenue is USD 270.3 million. Seasonality is retained rather than extrapolating Q2 strength in a straight line.

The portfolio uses calendar-quarter labels. For the near-term calibration, the issuer's closest reporting quarter is used as a proxy: its reported second quarter ended July 5, rather than June 30. These are calendar-labeled forecasts, not reconstructed daily calendar actuals. The funding roll begins at the actual reported July 5 balance and conservatively prorates its first modeled quarter; the small reporting-date mismatch is disclosed rather than presented as new economic growth.



Full-Year 2026 Outlook

All figures in $ million, unless specified	2026 Guidance	2025 Results	2025 Results ex CP*	Y/Y Change**	Y/Y Change** ex CP*
Revenue	**$1,130 – $1,150**	**$994**	**$982**	**+15%**	**+16%**
Adj. EBITDA Margin[1]	**29% – 31%**	**21.5%**	**20.7%**	**+850bps**	**+930bps**
Adj. Diluted EPS[1]	**$1.64 – $1.68**	**$1.02**	**$0.97**	**+63%**	**+71%**

* Excluding the one-time benefit from the commercial partnership with a medical lab automation channel partner (the "CP").
** At the midpoint of guidance.

[1]Cognex has provided the forward-looking non-GAAP measures of adjusted EBITDA margin and adjusted earnings per share (diluted), but cannot, without unreasonable effort, forecast such items to present or provide a reconciliation to corresponding forecasted GAAP measures. These include special items such as reorganization charges, acquisition and integration charges, and amortization of acquisition-related intangible assets, all of which are subject to limitations in predictability of timing, ultimate outcome and numerous conditions outside of Cognex's control. Additionally, these items are outside of Cognex's normal business operations and not used by management to assess Cognex's operating results. Cognex believes these limitations would result in a range of projected values so broad as to not be meaningful to investors. For these reasons, Cognex believes that the probable significance of such information is low. Information with respect to special items for certain historical periods is included in the section entitled "Reconciliation of Selected Items From GAAP to Non-GAAP". In 2025 the GAAP operating margin was 16.3% and GAAP earnings per share (diluted) were $0.68.

2026 EXPECTATIONS

1. Year-over-year revenue growth reflects contribution from broader Factory Automation end markets and Logistics.
2. Adjusted EBITDA margin expansion driven by revenue growth and cost reduction resulting in 87% revenue flow through at the midpoint.
3. Adjusted EPS growth driven by revenue growth and margin expansion.

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August 5 guidance predates the proposed acquisition: USD 1.13-1.15 billion core sales and 29-31% adjusted EBITDA margin. Valuation retains amortization, SBC, depreciation and retention expense.

Original Cognex Q2 2026 presentation, page 10. These are core-company adjusted guidance metrics, not a fully expensed post-acquisition profit forecast.

The comparison bridge matters. The issuer identifies USD 13 million of prior-year commercial-partnership benefit and USD 17 million of portfolio exits. A separate USD 7 million order shift moves revenue from Q3 into Q2 and does not increase annual sales. One transcript passage conflates the first two items into a USD 30 million commercial-partnership benefit; the model follows the primary slide and the Q1 explanation instead.

Baselining Revenue for Comparability

TOPIC *All figures in $ million*		Q3	Q4	Full Year
2025 GAAP Revenue as reported		**$277**	**$252**	**$994**
Completed portfolio optimization actions	*Y/Y headwind*	(5)	(5)	(17)
Consumer Electronics order timing shift (Q3 → Q2)	*Timing*	(7)	--	--
One-time commercial partnership benefit (Q3 2025)	*Y/Y headwind*	(13)	--	(13)
Revenue Baseline (for growth comparison)		**252**	**247**	**964**
2026 GAAP Revenue guidance range[1]		**$300 – 320**	**--**	**$1,130 – 1,150**
Implied Y/Y growth at the mid-point		*23%*	--	*18%*

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1) Q4 2026 GAAP revenue guidance to be provided in the Q3 quarterly earnings release.

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Primary comparison bridge: USD 13 million commercial partnership, USD 17 million portfolio exits, and aUSD 7 million Q3-to-Q2 order shift. The latter does not increase annual sales.

Original Cognex Q2 2026 presentation, page 11. Partnership, portfolio exits and intra-year timing are separate adjustments. They are not three recurring sources of incremental growth.

Management's approximately USD 35 million annualized cost-savings objective is already reflected in the exit cost structure. The model does not add it again or perpetuate an unusually high quarterly incremental margin. A disclosed approximately 75-basis-point Q3 memory-component gross-margin headwind is incorporated in the calibrated 2026 gross profit. Management expects pricing to offset component inflation in 2027; that recovery is inside the reference, not the additional value-pricing branch.

GAAP to Non-GAAP Tables

USD $ in 000s unless noted as per share

	Three-months Ended		Six-months Ended	
	July 5, 2026	June 29, 2025	**July 5, 2026**	June 29, 2025
Gross profit (GAAP)	**$ 205,773**	$ 167,876	**$ 396,712**	$ 312,199
Acquisition and integration costs	**218**	211	**434**	453
Amortization of acquisition-related intangible assets	**1,323**	1,382	**2,660**	2,720
Reorganization charges	**921**	—	**1,295**	86
Adjusted gross profit	**$ 208,235**	$ 169,469	**$ 401,101**	$ 315,458
GAAP gross margin	**70.6 %**	67.4 %	**70.9 %**	67.1 %
Adjusted gross margin	**71.5 %**	68.0 %	**71.7 %**	67.8 %
Operating expense (GAAP)	**$ 120,256**	$ 124,443	**$ 251,322**	$ 242,674
Acquisition and integration costs	**(15)**	(259)	**(30)**	(797)
Amortization of acquisition-related intangible assets	**(972)**	(1,296)	**(2,167)**	(2,586)
Reorganization charges	**(335)**	—	**(5,090)**	(1,622)
Adjusted operating expense	**$ 118,934**	$ 122,888	**$ 244,035**	$ 237,669
Operating income (GAAP)	**$ 85,517**	$ 43,433	**$ 145,390**	$ 69,525
Acquisition and integration costs	**233**	470	**464**	1,250
Amortization of acquisition-related intangible assets	**2,295**	2,678	**4,827**	5,306
Reorganization charges	**1,256**	—	**6,385**	1,708
Adjusted operating income	**$ 89,301**	$ 46,581	**$ 157,066**	$ 77,789
GAAP operating margin	**29.4 %**	17.4 %	**26.0 %**	14.9 %
Adjusted operating margin	**30.7 %**	18.7 %	**28.1 %**	16.7 %
Depreciation (adjusted for amounts included in Acquisition and integration costs)	**4,358**	5,095	**8,830**	10,178
Adjusted EBITDA	**$ 93,659**	$ 51,676	**$ 165,896**	$ 87,967
Adjusted EBITDA margin	**32.2 %**	20.7 %	**29.6 %**	18.9 %

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Amounts are USD thousands. H1 GAAP OP 145.390 million reconciles to adjusted EBITDA 165.896 million; SBC is already retained in adjusted figures and is not expensed a second time.

Original Cognex Q2 2026 presentation, page 13; amounts are USD thousands. The model carries ordinary depreciation, stock compensation and acquired-intangible amortization through operating profit.

Beyond 2026, the reference grows each paid application pool separately. Logistics and packaging grow 11% in 2027; consumer electronics 8%; automotive 3%; semiconductor OEM vision and optics 15%; other applications 5%; and the carved-out data-center pool 30%. Growth moderates subsequently. Product gross margins are analytical assumptions of 69-74%, not disclosed segment margins. Reference core operating expenses rise 6% annually from a USD 113 million quarterly starting point. Normal stock compensation, depreciation, existing amortization and a continuing restructuring reserve remain expensed.

First-half stock compensation was USD 23.145 million, including forfeiture-related credits. The forecast normalizes annual core stock compensation to USD 50 million in 2027, growing 6% thereafter, rather than extrapolating a one-time forfeiture benefit. Approximately 5% is embedded in product cost and the balance in operating expenses. SBC is added back solely when testing cash flow; it remains an economic expense in valuation and its share consequences remain in the denominator.

The model uses USD 5,000 per core invoice-equivalent to express the revenue bridge consistently. This is an analytical unit convention, not a claimed camera ASP or published shipment count. Actual products vary widely in price and content. The revenue and gross-profit tables are the controlling forecasts until real product-level units, mix and capacity are available.

Probability-weighted engineering and pricing branches

The reference already includes the cyclical recovery, current launches, data-center inspection and component-cost recovery. The following changes are incremental to that reference and begin in 2027. They are assumptions informed by company evidence, not company guidance.

Additional application adoption - 55%. A pool equal to 20% of reference logistics plus packaging sales receives 8% additional invoice-equivalent volume in 2027, 12% in 2028 and 15% in 2029-2030. A 72% incremental gross margin and 20% incremental selling, support and engineering expense produce 52% operating-profit capture. Existing new-customer counts do not justify scaling the whole company by the same rate. Qualification, deployment and acceptance govern recognition.

Value-based pricing beyond inflation reimbursement - 45%. A 25% pool of core sales receives a further 2% price level in 2027 and 3% from 2028 onward. These levels do not compound annually. An 80% incremental gross margin and 10% incremental commercial/support expense leave 70% operating-profit capture. The pool covers new or repriced business; it is not a surcharge on every committed order. Reference recovery of memory costs is already included, and no additional surcharge is applied to the adoption branch or RealSense.

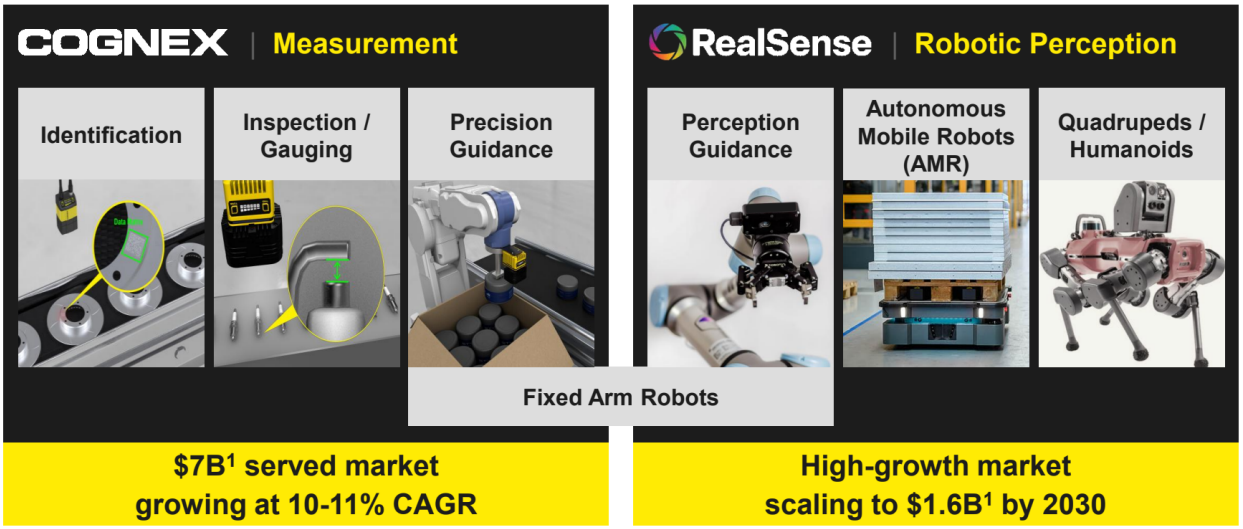
Core cycle and component-cost downside - 30%. Core revenue falls 8% below reference in 2027-2028 and 5% below reference thereafter. Lost sales remove their associated product gross profit, while only 15% of lost revenue is saved in variable operating expenses. Surviving revenue also carries a 1-point gross-margin cost in 2027-2028 and 0.5 point later, covering adverse mix, expedited components and competitive pressure. The positive adoption and pricing pools shrink with the same lost base. This avoids receiving full upside on sales that disappear in the adverse state.

The probabilities are explicit judgments about a bounded increment. Officially announced across-the-board price increases are not required before assigning some value to better applications, customer labor savings and line downtime avoided. Conversely, component shortages alone do not prove that Cognex has a lasting monopoly or that every customer will accept a premium. Management's three-to-four-month visibility limits confidence in the multi-year path.

What RealSense changes

The September 22 agreement is proposed, not closed, as of this review. RealSense originated at Intel and became independent in 2025. Its depth-perception cameras and software help robots and other machines perceive distance and navigate their environment. Cognex's historical precision-measurement applications and RealSense's streaming perception have different resolution, latency, frame-rate and integration priorities. The combination can broaden Cognex's customer problem set without making either architecture a universal replacement for the other.

Expansion into High-Growth Robotic Perception



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1) Source: Cognex estimates based on market research data.

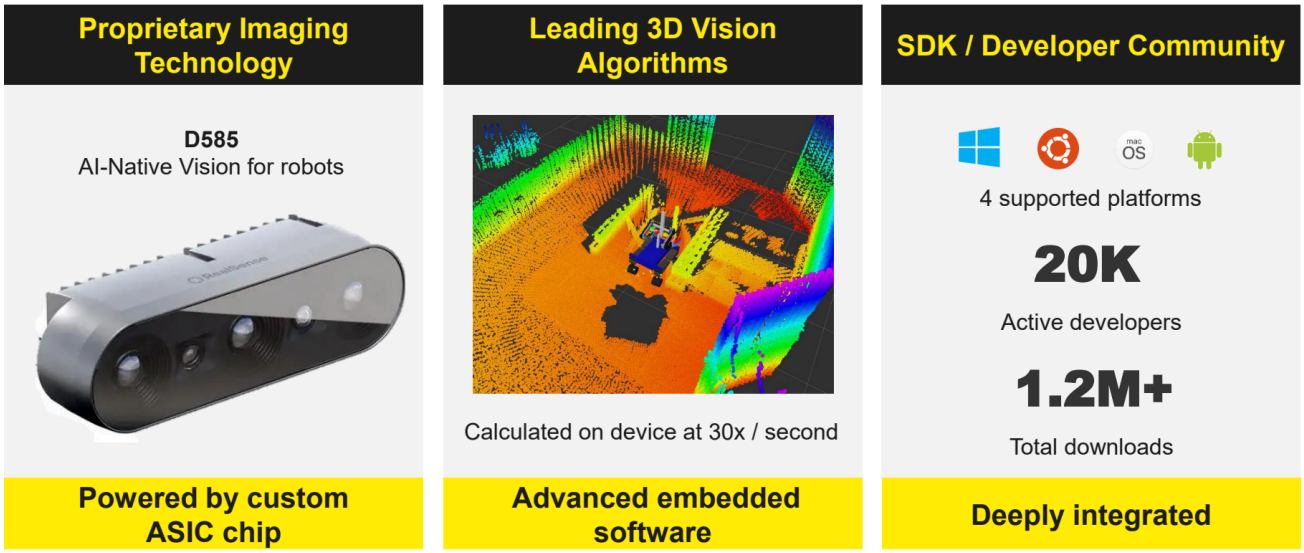
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Precision measurement and high-frame-rate navigation solve different tasks. The market sizes are issuer estimates, not signed customer revenue.

Original Cognex RealSense presentation, page 5. Market estimates describe an opportunity, not contracted Cognex revenue. Fixed industrial robots and autonomous mobile robots matter without assuming a humanoid boom.

RealSense reports more than one million cumulative cameras shipped, over 20,000 active developers and more than 1.2 million software downloads. These are installed-base and ecosystem indicators. They are not annual camera demand, paying-developer counts or recurring revenue. Its OEM-design business can scale through qualified designs, but OEM concentration, design cycles and alternative sensors remain material risks.

Defensible Leadership in 3D Robotic Perception



The D585 image illustrates on-device perception. 20,000 active developers and 1.2 million downloads are ecosystem measures; one million cumulative cameras shipped is not annual paid demand.

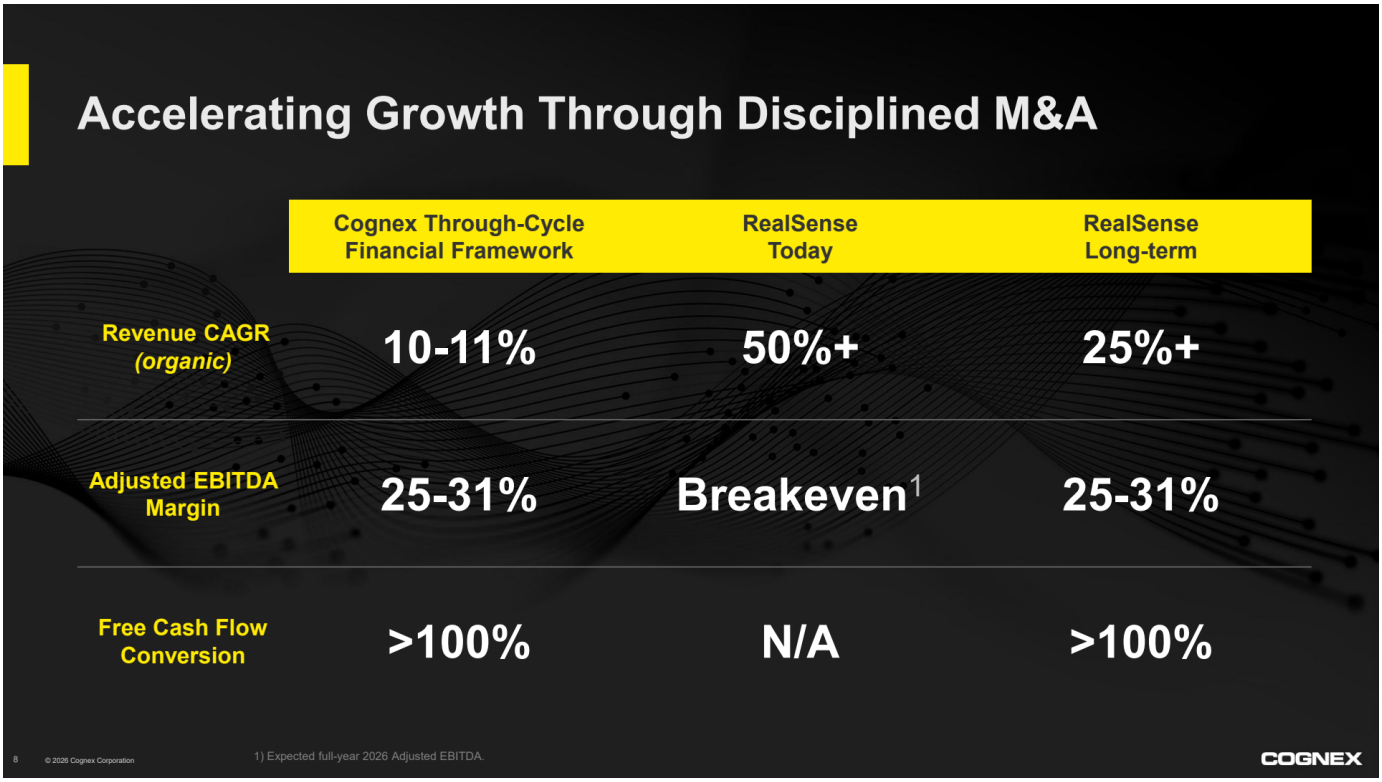
Original Cognex RealSense presentation, page 6. The processed-depth example and developer statistics do not establish a universal throughput or revenue benchmark.

The acquisition deck's 30-per-second processed-depth example should not be confused with every product mode. The current official D585 Pro page describes maximum depth throughput up to 90 frames per second, RGB throughput up to 60 frames per second and a specified 1280x960 mode, with stereo imaging, a global shutter and an IP65 enclosure. These are product specifications under stated configurations; they are not proof of the same frame rate at every resolution, full application latency or safety certification. Lighting, distance, reflectivity, occlusion and calibration still determine whether a depth estimate is usable.

LiDAR, structured illumination, stereo cameras and learned 2D approaches compete or combine depending on the environment and task. The value is a qualified, economical perception solution. Facial authentication will be separated before closing and is excluded here. Potential future deployment of OneVision models on RealSense hardware is not assigned an additional cross-selling revenue line before commercial evidence exists.

Acquisition economics and coherent states

The USD 500 million cash purchase price excludes target cash-retention awards of USD 56.5 million and approximately USD 50 million of restricted stock units. Retention is spread across three years, with terms and performance modifiers. The grant value and final legal share count remain uncertain. A planning reserve of USD 50 million divided by the presentation's USD 63.82 reference share price equals approximately 0.783 million shares. The model includes the full reserve from closing; this is neither an already issued share count nor a claim that the eventual award is fixed at that number.



RealSense 2026 adjusted EBITDA is expected around break-even. 25-31% is a long-term goal, before the acquisition retention expenses the issuer intends to exclude.

Original Cognex RealSense presentation, page 8. Approximate 2026 adjusted EBITDA break-even and a long-term 25-31% margin goal have different dates. The latter is not immediate GAAP profitability after retention and amortization.

The issuer expects 2026 RealSense sales of USD 80-90 million, growth above 50%, and approximately break-even adjusted EBITDA. The reference takes USD 85 million and a subsequent 25% annual sales-growth path. It models product gross margins rising from 53% in 2027 to 56% in 2030 and normal operating costs growing 12% annually. Normal engineering expense is calibrated so the 2026 starting business is approximately adjusted EBITDA break-even.

The purchase-price allocation is not available. The model explicitly assumes USD 300 million of amortizing acquired intangibles with a ten-year life, or USD 30 million annual amortization. Seventy percent enters cost of goods sold and 30% operating expenses. Cash retention of USD 56.5 million and stock compensation of USD 50 million are fully expensed over 12 quarters, alongside ordinary stock compensation and depreciation. A further USD 10 million transaction-cost assumption is expensed and paid. These are model assumptions requiring revision at closing; omitting them would overstate the value of a growth acquisition.



USD 500 million purchase is separate from USD 56.5 million target cash retention and approximate USD 50 million RSUs. Footnotes make grant-date economics conditional; all modeled costs are retained.

Original Cognex RealSense presentation, page 9. The USD 500 million purchase, cash retention and share awards are different claims. Presentation footnotes condition the grant economics on the eventual award terms and stock price.

Three deal outcomes are used: **80% timely closing in Q4 2026**, with the first full consolidated quarter in Q1 2027; **10% delayed closing in Q2 2027**, with consolidation from Q3; and **10% failure**, with no acquired operations or awards but the modeled transaction expense. This is a probability judgment, not a legal closing forecast.

A separate **35% acquired-business execution downside** applies only after completion. Relative to the 25% annual growth path, RealSense revenue is 20% lower in 2027, 25% lower in 2028 and 30% lower thereafter, with a further 3-point product gross-margin reduction. Normal operating costs, purchase amortization and committed retention remain. This captures slower OEM conversion and competing-camera pressure without inventing and then removing a humanoid-sales windfall.

The three deal outcomes and four binary engineering/risk events create 48 coherent operating states. Every state carries its own sales, costs, award shares, acquisition payment and cash roll. Its common-equity value is calculated first, then probability weighted. Dividing expected profit by an expected share count would miss the association between acquired profit, acquisition costs and dilution. Positive application gains and core demand weakness also interact on the same remaining sales base.

Cash, shares and funding capacity

At July 5, cash and cash equivalents were USD 302.521 million, current marketable securities USD 101.849 million and non-current marketable securities USD 350.643 million, a total of USD 755.013 million. The latter are available-for-sale securities measured at fair value, not restricted deposits. Non-current accounting classification does not make them cash, however: liquidation involves market, interest-rate and settlement risk. The securities portfolio totals USD 452.492 million and consists predominantly of corporate bonds. A 5% haircut to that portfolio removes approximately USD 22.625 million from opening value, roughly USD 0.13 per share.

This company's new valuation method credits only **fixed opening excess liquidity**: USD 755.013 million less a USD 200 million operating reserve. The completed-deal outcomes consume USD 500 million purchase cash and USD 10 million transaction cost from that credit. The failure outcome consumes only transaction costs. Acquisition cash is deducted even in an earlier ceiling window if that window includes the acquisition or acquired earnings; there is no free pre-closing purchase. Cash retention is subsequently expensed in operating profit and the funding roll. It is not silently included in the USD 500 million price.

Future retained cash is not added to the valuation. A separate funding roll tests whether the business can complete the transaction while paying capex, tax, dividends and retention. It includes an 18% positive-operating-profit cash-tax assumption, ordinary depreciation and SBC addbacks, a working-capital/cash-conversion reserve, core capex of 1.2% of revenue and acquired capex of 2%. These are assumptions, not a company cash-flow forecast. The first forecast quarter is prorated from the July 5 reported opening balance.

Across all 48 modeled states, minimum liquid assets remain approximately **USD 349.8 million**, above the USD 200 million reserve, without new debt or future option-exercise proceeds. This demonstrates feasibility under the chosen assumptions rather than immunity to an unmodeled draw. The model does not treat the remaining share-repurchase authorization as cash already spent. No executed post-July 5 amount was found in the reviewed sources; that reconciliation remains open. Any subsequently verified buyback requires both a cash reduction and the corresponding actual share reduction.

Reference diluted shares begin at 172 million and rise 0.20 million per quarter. The starting reserve is above 168.224 million issued shares at August 2 and the Q2 diluted average of 169.989 million; it accommodates outstanding awards, including reported anti-dilutive instruments, without claiming that dated issued shares equal a future diluted denominator. Each completion state adds the RealSense award reserve. Valuation uses the end of each four-quarter window. Leases are not deducted again as financial debt because rent is already retained in operating costs. Cognex reported no financial borrowing in the reviewed balance sheet.

What would change conviction

The most useful next evidence is conversion of OneVision and SLX use into paid production applications, actual price realization beyond component recovery, and continued high customer retention without excessive implementation expense. New customer counts alone are insufficient. In semiconductor equipment, the key evidence is qualified product content and shipped tools; in data centers, it is accepted inspection stations rather than a customer's total build budget.

For RealSense, closing documents, the purchase-price allocation, retained engineering headcount, legal share awards and the first full quarter of acquired revenue and margin are decisive. The existing-rooftop manufacturing capacity, outsourced camera supply and qualified design-win conversion supporting the modeled 2030 volume remain gaps. At a purely analytical USD 500 camera-equivalent, reference 2030 RealSense sales would imply roughly 415,000 equivalents; actual product ASPs and units may differ materially. This is a capacity question to verify, not a disclosed shipment forecast.

Downside evidence would include a sustained customer capex pause, loss of a concentrated account, deterioration in price realization, larger component prepayments, camera commoditization or lower RealSense revenue with retention costs unchanged. A rapid move toward open software and easily interchangeable hardware could weaken the installation economics despite a larger automation market. Conversely, independently confirmed deployment premiums and strong yield/customer-labor savings can justify revising the bounded price and application branches before an entire industry reports the same result.

The Base score is 16/20 for Core, 8/10 for Cycle, 7/10 for TAM, 4/5 for Catalysts, 7/10 for Macro and 4/5 for Alpha, less 3 points for Risk. That is 43/60, rounded to 72/100. Installed tools, image formation and qualification support the core; competition, short visibility and acquisition cost uncertainty prevent a higher classification. This analytical score is independent of valuation and of any portfolio override.

Source register and reporting limits

The full Q1 and Q2 management discussions, including questions and answers, and the September 22 acquisition discussion were reviewed alongside issuer filings and all 40 pages of the three source presentations. Ten original exhibits were inspected at readable resolution and retained above. Reported facts, company objectives and analytical allocations have different evidentiary status; the reference and active forecasts beyond guidance remain assumptions.

- Cognex 2025 Form 10-K, filed February 12, 2026: competition, geography, customer concentration and operating structure. - Q2 2026 Form 10-Q, August 6: reported financials, securities, awards, lease claims and issued shares. - Q1 presentation and full Q1 discussion, May 6-7: product architecture, value pricing and comparison adjustments. - Q2 presentation and full Q2 discussion, August 5-6; transcript published August 12: end markets, guidance, cost inflation and data-center scope. - RealSense transaction presentation, issuer announcement and full acquisition discussion, September 22: consideration, exclusions, retention and dated operating goals. - RealSense D585 Pro specifications and KEYENCE vision systems, reviewed October 7: product-mode boundaries and competitive evidence.

Revenue pools and operating sensitivity

The following are analytical reference allocations in USD millions. They reconcile to one company total; Cognex does not report these as separate financial segments. RealSense is excluded from this core table and enters only in completed-deal outcomes.

Paid application pool	2026	2027	2028	2030
Logistics barcode and vision	278.0	308.6	336.4	396.0
Packaging inspection	225.0	249.8	274.7	326.4
Consumer electronics excluding data centers	214.0	231.1	247.3	280.5
Automotive inspection	190.0	195.7	203.5	220.1
Semiconductor OEM vision and Moritex optics	147.0	169.0	191.0	235.3
Other industrial applications	56.0	58.8	61.7	68.1

Paid application pool	2026	2027	2028	2030
Data-center supply-chain inspection	30.0	39.0	48.8	67.3
Total core revenue	1,140.0	1,252.0	1,363.4	1,593.6

A one-percentage-point core product gross-margin shortfall reduces operating profit by 1% of the affected core revenue, before any cash tax, because the model does not assume an immediate matching reduction in engineering or selling expense. A 10% revenue shortfall is more costly than a 10% profit shortfall: at fixed product margins and operating costs it removes the lost sales' gross profit. The adverse modeled branch allows only a limited 15% variable-expense saving. These sensitivities are separate ways to examine the same earnings base and should not automatically be added together.

Quarterly funding bridge

All figures below are USD millions. Expected figures are probability weighted; the final column is the lowest closing balance among the 48 coherent states, rather than the lowest balance of an averaged company. Every state pays its own retention, acquisition cash, taxes and award-related costs. Financial debt remains zero in these modeled paths. Investment securities remain liquid but market-risk-bearing. No future retained cash is added to valuation.

Quarter	Expected CFO	Capex + dividends	Purchase cash	Expected liquid assets	Lowest state liquid assets
Q3 2026	77.1	17.8	0.0	814.3	814.3
Q4 2026	53.8	17.5	400.0	450.6	349.8
Q1 2027	69.3	18.7	0.0	501.3	392.3
Q2 2027	83.8	19.0	50.0	516.1	446.0
Q3 2027	94.0	19.2	0.0	590.9	509.0
Q4 2027	74.7	18.7	0.0	647.0	553.9
Q1 2028	81.4	19.6	0.0	708.7	603.2
Q2 2028	96.0	19.9	0.0	784.8	665.5
Q3 2028	107.0	20.1	0.0	871.7	738.0
Q4 2028	85.4	19.6	0.0	937.5	790.7
Q1 2029	93.8	20.6	0.0	1,010.7	851.9
Q2 2029	110.4	20.8	0.0	1,100.3	929.7
Q3 2029	122.4	21.1	0.0	1,201.6	1,018.8
Q4 2029	98.7	20.5	0.0	1,279.8	1,085.7
Q1 2030	108.1	21.5	0.0	1,366.4	1,160.7
Q2 2030	124.8	21.8	0.0	1,469.4	1,251.2
Q3 2030	138.1	22.1	0.0	1,585.4	1,353.9
Q4 2030	112.6	21.4	0.0	1,676.5	1,432.7

The transaction-cost assumption is included in operating profit and CFO, so it is not deducted again in the purchase-cash column. The latter contains the USD 500 million purchase only. The cash-retention expense is also already inside CFO. The USD 200 million reserve is tested against each state, not only the expectation. Unmodeled buybacks, changes in acquired working capital or lower liquidation proceeds require the funding roll to be repeated.

Valuation windows and state contributions

USD per share. Each column headed by a deal outcome is its already probability-weighted contribution after evaluating the four operating-event branches and that outcome's own shares and cash claims. The three contributions sum to the active ceiling. The reference is the distinct no-deal, all-incremental-events-off company; it incurs no attempted-acquisition cost. All figures use 20x fully expensed operating profit for the labeled quarter plus the following three quarters.

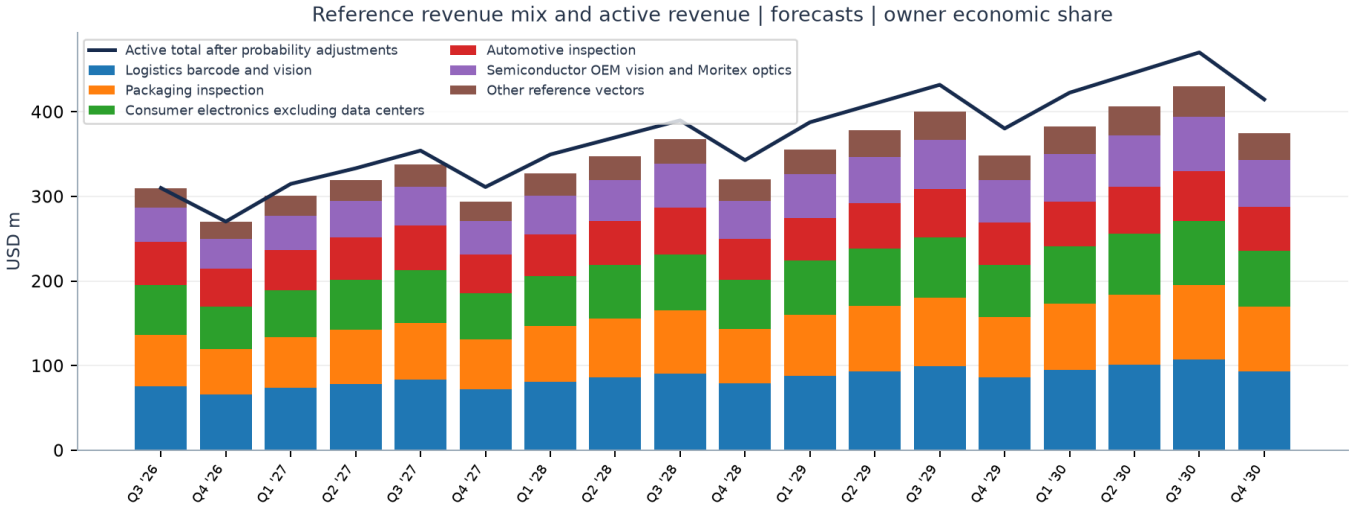
Ceiling quarter	No-deal reference	Timely 80% contribution	Delayed 10% contribution	Failure 10% contribution	Active ceiling
Q3 2026	44.83	28.71	3.96	4.27	36.94
Q4 2026	47.33	28.82	3.98	4.47	37.27
Q1 2027	49.86	29.94	4.00	4.80	38.73
Q2 2027	51.20	31.19	3.97	4.93	40.09
Q3 2027	52.67	32.56	4.07	5.08	41.71
Q4 2027	54.28	34.03	4.25	5.25	43.53
Q1 2028	55.56	35.24	4.41	5.38	45.02
Q2 2028	56.96	36.79	4.60	5.54	46.93
Q3 2028	58.51	38.46	4.81	5.73	49.00
Q4 2028	60.20	40.26	5.03	5.92	51.22
Q1 2029	61.54	41.76	5.22	6.08	53.06
Q2 2029	62.74	43.85	5.38	6.20	55.43
Q3 2029	64.08	46.04	5.55	6.34	57.93
Q4 2029	65.54	48.34	5.84	6.48	60.66
Q1 2030	66.69	50.38	6.10	6.60	63.07

A lower immediate value in the completed-acquisition outcomes is possible even if RealSense is strategically useful: purchase cash, fully expensed retention and acquired-intangible amortization can outweigh its early profit. The model does not force an announced acquisition to be accretive. Later ceiling changes also reflect the expiry of the finite retention expense, continuing ordinary costs and end-window dilution.

Score arithmetic

Category	Points	Evidence / treatment
Core thesis	16	16/20: qualified vision tools, image formation, OEM designs and installed workflows protect value; capable rivals and open AI prevent monopoly economics.
Cycle	8	8/10: broad 2026 recovery and profitable growth are verified; three-to-four-month visibility and customer capex cycles remain.
TAM	7	7/10: paid inspection and robotic-perception units expand, but neither global automation capex nor humanoid shipments directly become Cognex sales.
Catalysts	4	4/5: OneVision adoption, new embedded systems, logistics attachments and conditional RealSense closing.
Macro	7	7/10: labor scarcity and quality requirements support automation, offset by China competition, tariffs and cyclical capital budgets.
Alpha	4	4/5: separates precision measurement from navigation and models acquisition retention, amortization, shares and liquidity together.
Risk classes	-3	-3: short-cycle demand, concentrated customers, component allocation and acquisition economics remain material.
Total	43/60	Rounded Base 72. Both artifact and new workbook row use this score; this is not a manual override.

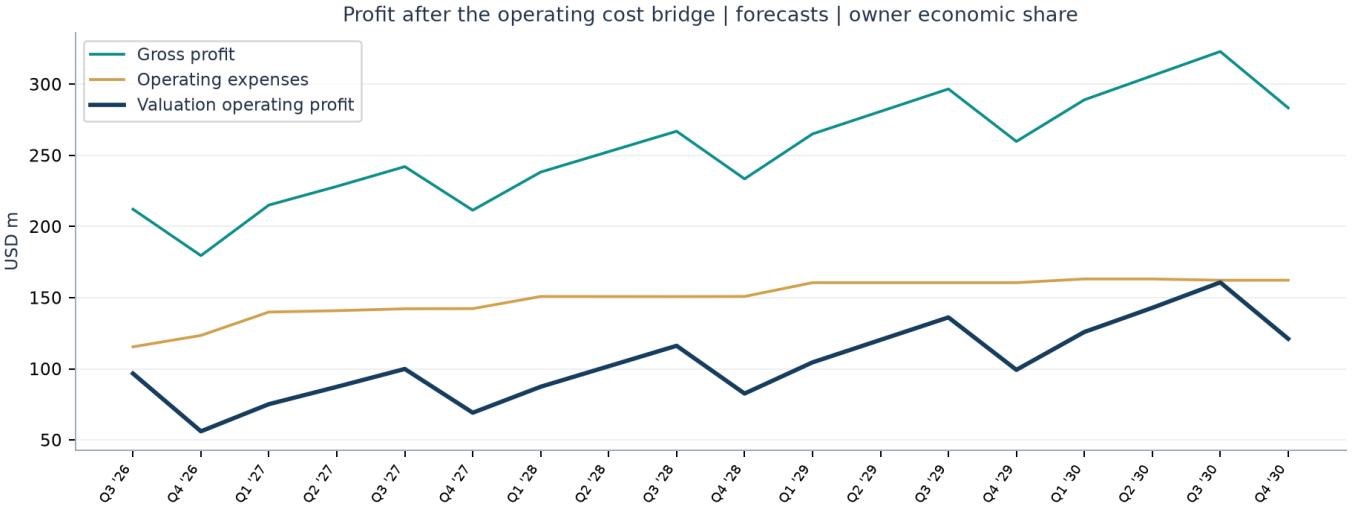
Revenue and business mix



Reference business vectors sum to the reference; the active total adds only probability-weighted net changes. Product allocations may be analytical proxies where issuers do not report product margins.

Quarter	Reference revenue	Weighted net change	Active revenue
Q3 2026	310.0	0.0	310.0
Q4 2026	270.3	0.0	270.3
Q1 2027	300.5	14.4	314.8
Q2 2027	319.3	14.0	333.3
Q3 2027	338.0	16.2	354.2
Q4 2027	294.2	16.9	311.2
Q1 2028	327.2	22.4	349.6
Q2 2028	347.7	22.1	369.7
Q3 2028	368.1	21.8	389.9
Q4 2028	320.4	22.5	342.9
Q1 2029	355.8	31.9	387.7
Q2 2029	378.0	31.8	409.8
Q3 2029	400.2	31.7	431.9
Q4 2029	348.4	31.9	380.3
Q1 2030	382.5	40.1	422.6
Q2 2030	406.4	40.0	446.4
Q3 2030	430.3	39.9	470.2
Q4 2030	374.5	40.2	414.7

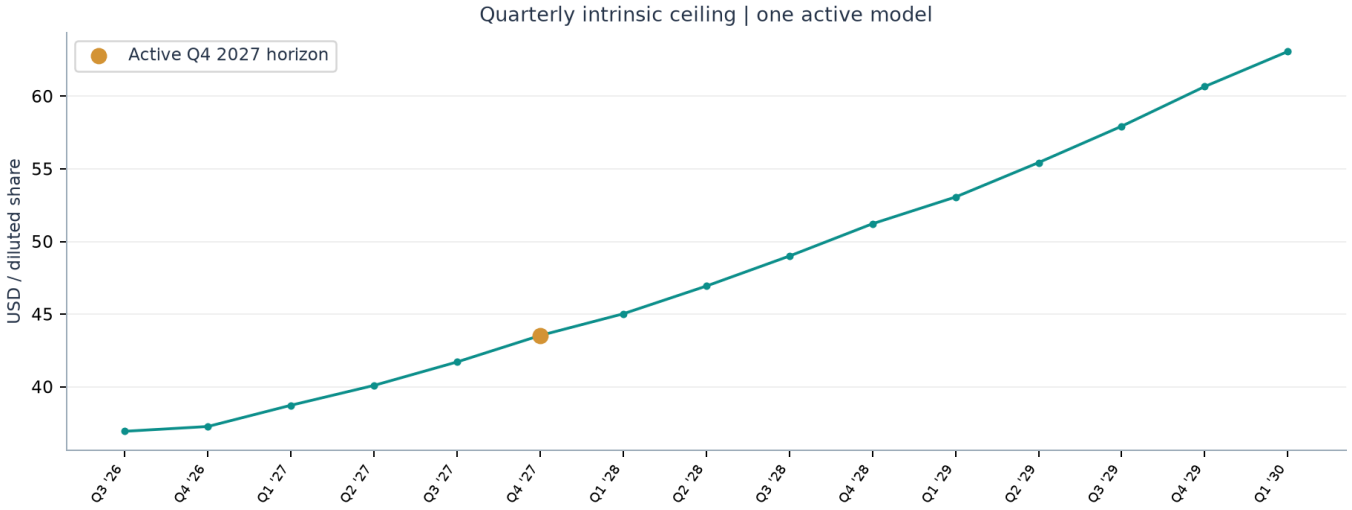
Gross profit, operating expenses and earnings



Operating profit equals gross profit less operating expenses and any separately identified attribution reserve. Figures are forecasts, not reported historical quarterly outcomes.

Quarter	Revenue	GP	Opex	Valuation OP	OP margin
Q3 2026	310.0	212.2	115.5	96.7	31.2%
Q4 2026	270.3	179.6	123.5	56.1	20.8%
Q1 2027	314.8	215.1	139.9	75.2	23.9%
Q2 2027	333.3	228.2	140.9	87.3	26.2%
Q3 2027	354.2	242.1	142.2	99.9	28.2%
Q4 2027	311.2	211.5	142.3	69.2	22.2%
Q1 2028	349.6	238.3	150.9	87.4	25.0%
Q2 2028	369.7	252.6	150.8	101.8	27.5%
Q3 2028	389.9	267.0	150.8	116.2	29.8%
Q4 2028	342.9	233.5	150.9	82.6	24.1%
Q1 2029	387.7	265.1	160.5	104.5	27.0%
Q2 2029	409.8	280.9	160.5	120.3	29.4%
Q3 2029	431.9	296.7	160.5	136.1	31.5%
Q4 2029	380.3	259.8	160.5	99.3	26.1%
Q1 2030	422.6	289.0	163.2	125.9	29.8%
Q2 2030	446.4	306.0	163.1	142.9	32.0%
Q3 2030	470.2	323.0	162.2	160.7	34.2%
Q4 2030	414.7	283.4	162.3	121.1	29.2%

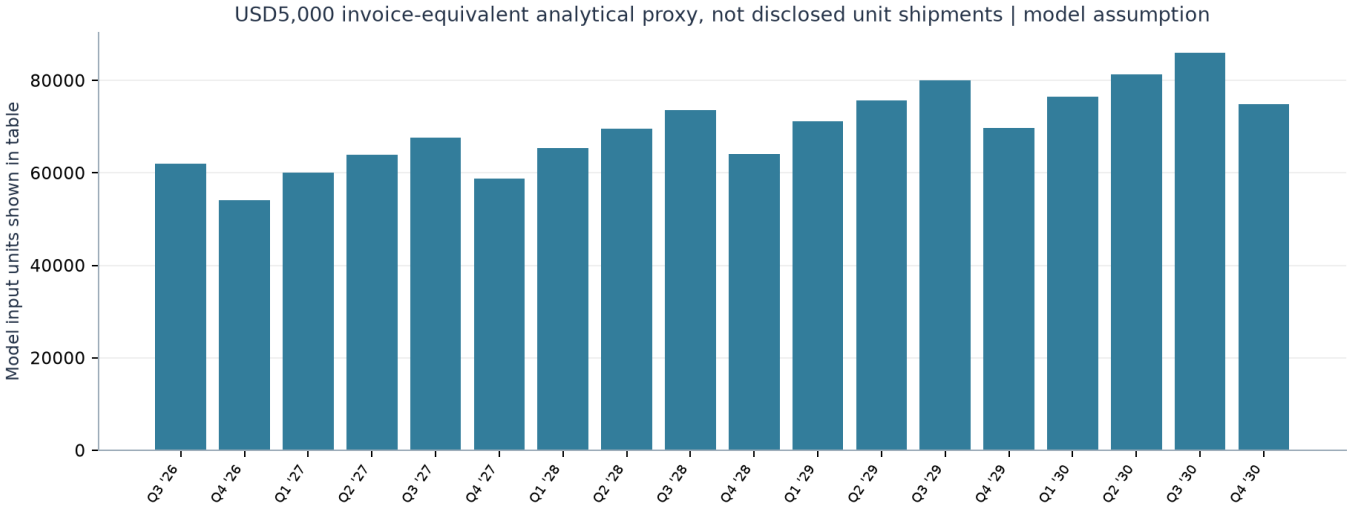
The complete ceiling path



Quarter Q prices Q plus the next three calendar quarters. All fifteen ceilings link to the same operating rows used in the workbook. Outer quarters have lower evidence visibility.

Quarter	Four-quarter earnings window	F4Q OP	Multiple	Shares	Ceiling
Q3 2026	Q3 2026 to Q2 2027	315.3	20x	173.305	36.94
Q4 2026	Q4 2026 to Q3 2027	318.5	20x	173.505	37.27
Q1 2027	Q1 2027 to Q4 2027	331.6	20x	173.705	38.73
Q2 2027	Q2 2027 to Q1 2028	343.8	20x	173.905	40.09
Q3 2027	Q3 2027 to Q2 2028	358.3	20x	174.105	41.71
Q4 2027	Q4 2027 to Q3 2028	374.6	20x	174.305	43.53
Q1 2028	Q1 2028 to Q4 2028	388.1	20x	174.505	45.02
Q2 2028	Q2 2028 to Q1 2029	405.2	20x	174.705	46.93
Q3 2028	Q3 2028 to Q2 2029	423.7	20x	174.905	49.00
Q4 2028	Q4 2028 to Q3 2029	443.7	20x	175.105	51.22
Q1 2029	Q1 2029 to Q4 2029	460.3	20x	175.305	53.06
Q2 2029	Q2 2029 to Q1 2030	481.6	20x	175.505	55.43
Q3 2029	Q3 2029 to Q2 2030	504.2	20x	175.705	57.93
Q4 2029	Q4 2029 to Q3 2030	528.8	20x	175.905	60.66
Q1 2030	Q1 2030 to Q4 2030	550.6	20x	176.105	63.07

The operating constraint behind the forecast



The driver series is a stated model assumption, not a disclosed commitment. It must be evaluated with the qualification, customer capture, cost and overlap discussion.

Quarter	USD5,000 invoice-equivalent analytical proxy, not disclosed unit shipments	Status
Q3 2026	62,000.000	Calendar-quarter analytical forecast; July5 reported balance opening
Q4 2026	54,060.000	Calendar-quarter analytical forecast; July5 reported balance opening
Q1 2027	60,096.000	Calendar-quarter analytical forecast; July5 reported balance opening
Q2 2027	63,852.000	Calendar-quarter analytical forecast; July5 reported balance opening
Q3 2027	67,608.000	Calendar-quarter analytical forecast; July5 reported balance opening
Q4 2027	58,844.000	Calendar-quarter analytical forecast; July5 reported balance opening
Q1 2028	65,444.165	Calendar-quarter analytical forecast; July5 reported balance opening
Q2 2028	69,534.425	Calendar-quarter analytical forecast; July5 reported balance opening
Q3 2028	73,624.685	Calendar-quarter analytical forecast; July5 reported balance opening
Q4 2028	64,080.745	Calendar-quarter analytical forecast; July5 reported balance opening
Q1 2029	71,154.071	Calendar-quarter analytical forecast; July5 reported balance opening
Q2 2029	75,601.201	Calendar-quarter analytical forecast; July5 reported balance opening

Quarter	USD5,000 invoice-equivalent analytical proxy, not disclosed unit shipments	Status
Q3 2029	80,048.330	Calendar-quarter analytical forecast; July5 reported balance opening
Q4 2029	69,671.695	Calendar-quarter analytical forecast; July5 reported balance opening
Q1 2030	76,494.568	Calendar-quarter analytical forecast; July5 reported balance opening
Q2 2030	81,275.479	Calendar-quarter analytical forecast; July5 reported balance opening
Q3 2030	86,056.389	Calendar-quarter analytical forecast; July5 reported balance opening
Q4 2030	74,900.931	Calendar-quarter analytical forecast; July5 reported balance opening

Quarterly revenue-vector derivation

All rows below are reference inputs. Gross profit = vector revenue x the stated gross margin, except explicitly labeled zero-sales cost credits. The next table applies the separately weighted evidence deltas. A product-level proxy is a model assumption even when the group total is calibrated to guidance.

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2026	Logistics barcode and vision	75.6	66.4%	50.2
Q3 2026	Packaging inspection	61.2	70.3%	43.0
Q3 2026	Consumer electronics excluding data centers	58.2	70.3%	40.9
Q3 2026	Automotive inspection	51.7	70.3%	36.3
Q3 2026	Semiconductor OEM vision and Moritex optics	40.0	67.4%	26.9
Q3 2026	Other industrial applications	15.2	66.4%	10.1
Q3 2026	Data-center supply-chain inspection	8.2	71.3%	5.8
Q3 2026	Existing acquired technology amortization	0.0	0.0%	-1.2
Q4 2026	Logistics barcode and vision	65.9	64.6%	42.6
Q4 2026	Packaging inspection	53.3	68.3%	36.4
Q4 2026	Consumer electronics excluding data centers	50.7	68.3%	34.7
Q4 2026	Automotive inspection	45.0	68.3%	30.8
Q4 2026	Semiconductor OEM vision and Moritex optics	34.9	65.5%	22.8
Q4 2026	Other industrial applications	13.3	64.6%	8.6
Q4 2026	Data-center supply-chain inspection	7.1	69.2%	4.9
Q4 2026	Existing acquired technology amortization	0.0	0.0%	-1.2
Q1 2027	Logistics barcode and vision	74.1	69.0%	51.1
Q1 2027	Packaging inspection	59.9	73.0%	43.8
Q1 2027	Consumer electronics excluding data centers	55.5	73.0%	40.5
Q1 2027	Automotive inspection	47.0	73.0%	34.3
Q1 2027	Semiconductor OEM vision and Moritex optics	40.6	70.0%	28.4
Q1 2027	Other industrial applications	14.1	69.0%	9.7
Q1 2027	Data-center supply-chain inspection	9.4	74.0%	6.9
Q1 2027	Existing acquired technology amortization	0.0	0.0%	-1.1
Q2 2027	Logistics barcode and vision	78.7	69.0%	54.3

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q2 2027	Packaging inspection	63.7	73.0%	46.5
Q2 2027	Consumer electronics excluding data centers	58.9	73.0%	43.0
Q2 2027	Automotive inspection	49.9	73.0%	36.4
Q2 2027	Semiconductor OEM vision and Moritex optics	43.1	70.0%	30.2
Q2 2027	Other industrial applications	15.0	69.0%	10.3
Q2 2027	Data-center supply-chain inspection	9.9	74.0%	7.4
Q2 2027	Existing acquired technology amortization	0.0	0.0%	-1.1
Q3 2027	Logistics barcode and vision	83.3	69.0%	57.5
Q3 2027	Packaging inspection	67.4	73.0%	49.2
Q3 2027	Consumer electronics excluding data centers	62.4	73.0%	45.6
Q3 2027	Automotive inspection	52.8	73.0%	38.6
Q3 2027	Semiconductor OEM vision and Moritex optics	45.6	70.0%	32.0
Q3 2027	Other industrial applications	15.9	69.0%	11.0
Q3 2027	Data-center supply-chain inspection	10.5	74.0%	7.8
Q3 2027	Existing acquired technology amortization	0.0	0.0%	-1.1
Q4 2027	Logistics barcode and vision	72.5	69.0%	50.0
Q4 2027	Packaging inspection	58.7	73.0%	42.8
Q4 2027	Consumer electronics excluding data centers	54.3	73.0%	39.6
Q4 2027	Automotive inspection	46.0	73.0%	33.6
Q4 2027	Semiconductor OEM vision and Moritex optics	39.7	70.0%	27.8
Q4 2027	Other industrial applications	13.8	69.0%	9.5
Q4 2027	Data-center supply-chain inspection	9.2	74.0%	6.8
Q4 2027	Existing acquired technology amortization	0.0	0.0%	-1.1
Q1 2028	Logistics barcode and vision	80.7	69.0%	55.7
Q1 2028	Packaging inspection	65.9	73.0%	48.1
Q1 2028	Consumer electronics excluding data centers	59.4	73.0%	43.3
Q1 2028	Automotive inspection	48.8	73.0%	35.7

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q1 2028	Semiconductor OEM vision and Moritex optics	45.8	70.0%	32.1
Q1 2028	Other industrial applications	14.8	69.0%	10.2
Q1 2028	Data-center supply-chain inspection	11.7	74.0%	8.7
Q1 2028	Existing acquired technology amortization	0.0	0.0%	-1.0
Q2 2028	Logistics barcode and vision	85.8	69.0%	59.2
Q2 2028	Packaging inspection	70.1	73.0%	51.1
Q2 2028	Consumer electronics excluding data centers	63.1	73.0%	46.0
Q2 2028	Automotive inspection	51.9	73.0%	37.9
Q2 2028	Semiconductor OEM vision and Moritex optics	48.7	70.0%	34.1
Q2 2028	Other industrial applications	15.7	69.0%	10.9
Q2 2028	Data-center supply-chain inspection	12.4	74.0%	9.2
Q2 2028	Existing acquired technology amortization	0.0	0.0%	-1.0
Q3 2028	Logistics barcode and vision	90.8	69.0%	62.7
Q3 2028	Packaging inspection	74.2	73.0%	54.1
Q3 2028	Consumer electronics excluding data centers	66.8	73.0%	48.7
Q3 2028	Automotive inspection	55.0	73.0%	40.1
Q3 2028	Semiconductor OEM vision and Moritex optics	51.6	70.0%	36.1
Q3 2028	Other industrial applications	16.7	69.0%	11.5
Q3 2028	Data-center supply-chain inspection	13.2	74.0%	9.7
Q3 2028	Existing acquired technology amortization	0.0	0.0%	-1.0
Q4 2028	Logistics barcode and vision	79.0	69.0%	54.5
Q4 2028	Packaging inspection	64.6	73.0%	47.1
Q4 2028	Consumer electronics excluding data centers	58.1	73.0%	42.4
Q4 2028	Automotive inspection	47.8	73.0%	34.9
Q4 2028	Semiconductor OEM vision and Moritex optics	44.9	70.0%	31.4
Q4 2028	Other industrial applications	14.5	69.0%	10.0
Q4 2028	Data-center supply-chain inspection	11.5	74.0%	8.5
Q4 2028	Existing acquired technology amortization	0.0	0.0%	-1.0

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q1 2029	Logistics barcode and vision	88.0	69.0%	60.7
Q1 2029	Packaging inspection	72.5	73.0%	52.9
Q1 2029	Consumer electronics excluding data centers	63.5	73.0%	46.4
Q1 2029	Automotive inspection	50.8	73.0%	37.1
Q1 2029	Semiconductor OEM vision and Moritex optics	51.3	70.0%	35.9
Q1 2029	Other industrial applications	15.6	69.0%	10.7
Q1 2029	Data-center supply-chain inspection	14.0	74.0%	10.4
Q1 2029	Existing acquired technology amortization	0.0	0.0%	-1.0
Q2 2029	Logistics barcode and vision	93.5	69.0%	64.5
Q2 2029	Packaging inspection	77.1	73.0%	56.3
Q2 2029	Consumer electronics excluding data centers	67.5	73.0%	49.3
Q2 2029	Automotive inspection	54.0	73.0%	39.4
Q2 2029	Semiconductor OEM vision and Moritex optics	54.6	70.0%	38.2
Q2 2029	Other industrial applications	16.5	69.0%	11.4
Q2 2029	Data-center supply-chain inspection	14.9	74.0%	11.0
Q2 2029	Existing acquired technology amortization	0.0	0.0%	-1.0
Q3 2029	Logistics barcode and vision	99.0	69.0%	68.3
Q3 2029	Packaging inspection	81.6	73.0%	59.6
Q3 2029	Consumer electronics excluding data centers	71.4	73.0%	52.2
Q3 2029	Automotive inspection	57.2	73.0%	41.7
Q3 2029	Semiconductor OEM vision and Moritex optics	57.8	70.0%	40.4
Q3 2029	Other industrial applications	17.5	69.0%	12.1
Q3 2029	Data-center supply-chain inspection	15.8	74.0%	11.7
Q3 2029	Existing acquired technology amortization	0.0	0.0%	-1.0
Q4 2029	Logistics barcode and vision	86.2	69.0%	59.4
Q4 2029	Packaging inspection	71.0	73.0%	51.8
Q4 2029	Consumer electronics excluding data centers	62.2	73.0%	45.4
Q4 2029	Automotive inspection	49.7	73.0%	36.3

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q4 2029	Semiconductor OEM vision and Moritex optics	50.3	70.0%	35.2
Q4 2029	Other industrial applications	15.2	69.0%	10.5
Q4 2029	Data-center supply-chain inspection	13.7	74.0%	10.2
Q4 2029	Existing acquired technology amortization	0.0	0.0%	-1.0
Q1 2030	Logistics barcode and vision	95.0	69.0%	65.6
Q1 2030	Packaging inspection	78.3	73.0%	57.2
Q1 2030	Consumer electronics excluding data centers	67.3	73.0%	49.1
Q1 2030	Automotive inspection	52.8	73.0%	38.6
Q1 2030	Semiconductor OEM vision and Moritex optics	56.5	70.0%	39.5
Q1 2030	Other industrial applications	16.3	69.0%	11.3
Q1 2030	Data-center supply-chain inspection	16.1	74.0%	11.9
Q1 2030	Existing acquired technology amortization	0.0	0.0%	-0.9
Q2 2030	Logistics barcode and vision	101.0	69.0%	69.7
Q2 2030	Packaging inspection	83.2	73.0%	60.8
Q2 2030	Consumer electronics excluding data centers	71.5	73.0%	52.2
Q2 2030	Automotive inspection	56.1	73.0%	41.0
Q2 2030	Semiconductor OEM vision and Moritex optics	60.0	70.0%	42.0
Q2 2030	Other industrial applications	17.4	69.0%	12.0
Q2 2030	Data-center supply-chain inspection	17.2	74.0%	12.7
Q2 2030	Existing acquired technology amortization	0.0	0.0%	-0.9
Q3 2030	Logistics barcode and vision	106.9	69.0%	73.8
Q3 2030	Packaging inspection	88.1	73.0%	64.3
Q3 2030	Consumer electronics excluding data centers	75.7	73.0%	55.3
Q3 2030	Automotive inspection	59.4	73.0%	43.4
Q3 2030	Semiconductor OEM vision and Moritex optics	63.5	70.0%	44.5
Q3 2030	Other industrial applications	18.4	69.0%	12.7
Q3 2030	Data-center supply-chain inspection	18.2	74.0%	13.4
Q3 2030	Existing acquired technology amortization	0.0	0.0%	-0.9

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q4 2030	Logistics barcode and vision	93.0	69.0%	64.2
Q4 2030	Packaging inspection	76.7	73.0%	56.0
Q4 2030	Consumer electronics excluding data centers	65.9	73.0%	48.1
Q4 2030	Automotive inspection	51.7	73.0%	37.8
Q4 2030	Semiconductor OEM vision and Moritex optics	55.3	70.0%	38.7
Q4 2030	Other industrial applications	16.0	69.0%	11.0
Q4 2030	Data-center supply-chain inspection	15.8	74.0%	11.7
Q4 2030	Existing acquired technology amortization	0.0	0.0%	-0.9

Operating-cost derivation

Quarter	R&D cash and ordinary depreciation	Selling, support and administration cash and ordinary depreciation	Operating stock compensation	Existing acquired customer intangible amortization	Continuing restructuring and integration reserve
Q3 2026	27.7	74.9	11.4	1.0	0.5
Q4 2026	27.4	74.2	11.4	1.0	0.5
Q1 2027	29.1	78.8	11.9	0.9	1.0
Q2 2027	29.1	78.8	11.9	0.9	1.0
Q3 2027	29.1	78.8	11.9	0.9	1.0
Q4 2027	29.1	78.8	11.9	0.9	1.0
Q1 2028	30.9	83.5	12.6	0.8	1.0
Q2 2028	30.9	83.5	12.6	0.8	1.0
Q3 2028	30.9	83.5	12.6	0.8	1.0
Q4 2028	30.9	83.5	12.6	0.8	1.0
Q1 2029	32.7	88.5	13.3	0.8	1.0
Q2 2029	32.7	88.5	13.3	0.8	1.0
Q3 2029	32.7	88.5	13.3	0.8	1.0
Q4 2029	32.7	88.5	13.3	0.8	1.0
Q1 2030	34.7	93.8	14.1	0.7	1.0
Q2 2030	34.7	93.8	14.1	0.7	1.0
Q3 2030	34.7	93.8	14.1	0.7	1.0
Q4 2030	34.7	93.8	14.1	0.7	1.0

The categories reconcile to reference operating expenses. Incremental costs below are applied once; depreciation or stock compensation already included in a reference category is not added again.

Probability-weighted financial bridge

Active = reference + sum(probability x bounded conditional delta). Negative deltas are retained. A branch with zero near-term increment remains zero only for the timing or overlap reasons documented in the evidence register.

For readability, the table groups the 48 coherent paths into three transaction outcomes. Each conditional delta averages the sixteen engineering-event combinations within that transaction outcome. The original probabilities, interactions and all state calculations remain explicit in the linked model; this grouping preserves every quarterly revenue, gross-profit and expense total.

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2026	timely deal outcome	80%	0.0	0.0	0.0	0.0
Q3 2026	delayed deal outcome	10%	0.0	0.0	0.0	0.0
Q3 2026	failure deal outcome	10%	0.0	0.0	0.0	0.0
Q4 2026	timely deal outcome	80%	0.0	0.0	10.0	-8.0
Q4 2026	delayed deal outcome	10%	0.0	0.0	0.0	0.0
Q4 2026	failure deal outcome	10%	0.0	0.0	10.0	-1.0
Q1 2027	timely deal outcome	80%	19.3	3.0	23.0	-16.0
Q1 2027	delayed deal outcome	10%	-5.4	-4.6	-0.8	-0.4
Q1 2027	failure deal outcome	10%	-5.4	-4.6	-0.8	-0.4
Q2 2027	timely deal outcome	80%	19.0	2.7	22.9	-16.2
Q2 2027	delayed deal outcome	10%	-5.7	-4.9	9.2	-1.4
Q2 2027	failure deal outcome	10%	-5.7	-4.9	-0.8	-0.4
Q3 2027	timely deal outcome	80%	18.6	2.4	22.9	-16.4
Q3 2027	delayed deal outcome	10%	18.6	2.4	22.9	-2.0
Q3 2027	failure deal outcome	10%	-6.1	-5.2	-0.9	-0.4
Q4 2027	timely deal outcome	80%	19.4	3.1	23.0	-15.9
Q4 2027	delayed deal outcome	10%	19.4	3.1	23.0	-2.0
Q4 2027	failure deal outcome	10%	-5.3	-4.5	-0.8	-0.4
Q1 2028	timely deal outcome	80%	25.4	6.6	24.6	-14.4
Q1 2028	delayed deal outcome	10%	25.4	6.6	24.6	-1.8
Q1 2028	failure deal outcome	10%	-4.9	-4.3	-0.7	-0.4
Q2 2028	timely deal outcome	80%	25.1	6.3	24.6	-14.6
Q2 2028	delayed deal outcome	10%	25.1	6.3	24.6	-1.8
Q2 2028	failure deal outcome	10%	-5.2	-4.6	-0.7	-0.4
Q3 2028	timely deal outcome	80%	24.8	6.0	24.5	-14.8
Q3 2028	delayed deal outcome	10%	24.8	6.0	24.5	-1.8

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2028	failure deal outcome	10%	-5.5	-4.8	-0.8	-0.4
Q4 2028	timely deal outcome	80%	25.5	6.6	24.6	-14.4
Q4 2028	delayed deal outcome	10%	25.5	6.6	24.6	-1.8
Q4 2028	failure deal outcome	10%	-4.8	-4.2	-0.7	-0.4
Q1 2029	timely deal outcome	80%	35.6	13.4	26.8	-10.8
Q1 2029	delayed deal outcome	10%	35.6	13.4	26.8	-1.3
Q1 2029	failure deal outcome	10%	-1.5	-1.5	-0.2	-0.1
Q2 2029	timely deal outcome	80%	35.5	13.3	26.8	-10.8
Q2 2029	delayed deal outcome	10%	35.5	13.3	26.8	-1.4
Q2 2029	failure deal outcome	10%	-1.6	-1.6	-0.2	-0.1
Q3 2029	timely deal outcome	80%	35.4	13.2	26.8	-10.9
Q3 2029	delayed deal outcome	10%	35.4	13.2	26.8	-1.4
Q3 2029	failure deal outcome	10%	-1.7	-1.7	-0.2	-0.2
Q4 2029	timely deal outcome	80%	35.6	13.4	26.8	-10.7
Q4 2029	delayed deal outcome	10%	35.6	13.4	26.8	-1.3
Q4 2029	failure deal outcome	10%	-1.5	-1.5	-0.2	-0.1
Q1 2030	timely deal outcome	80%	44.8	18.8	19.9	-0.9
Q1 2030	delayed deal outcome	10%	44.8	18.8	28.7	-1.0
Q1 2030	failure deal outcome	10%	-1.6	-1.6	-0.2	-0.1
Q2 2030	timely deal outcome	80%	44.7	18.7	19.9	-0.9
Q2 2030	delayed deal outcome	10%	44.7	18.7	28.7	-1.0
Q2 2030	failure deal outcome	10%	-1.8	-1.7	-0.2	-0.2
Q3 2030	timely deal outcome	80%	44.6	18.6	19.8	-1.0
Q3 2030	delayed deal outcome	10%	44.6	18.6	19.8	-0.1
Q3 2030	failure deal outcome	10%	-1.9	-1.8	-0.2	-0.2
Q4 2030	timely deal outcome	80%	44.8	18.8	19.9	-0.8
Q4 2030	delayed deal outcome	10%	44.8	18.8	19.9	-0.1
Q4 2030	failure deal outcome	10%	-1.6	-1.6	-0.2	-0.1

Claim decisions, overlap and triggers

REALSENSE_TRANSACTION

p: 0.9

condition: The signed acquisition closes, with 80% probability in Q4 2026 and 10% in Q2 2027; 10% fails.

Conditional: USD 500 million purchase and USD 10 million modeled transaction costs. Operating consolidation starts the following full quarter. Reference RealSense sales start from USD 85 million in 2026 and grow 25% annually; its separate 35% execution-risk branch reduces that path.

cost: Full USD 56.5 million target cash retention and USD 50 million RSU expense over 12 quarters; USD 30 million annual modeled purchase amortization; normal engineering and support costs grow 12% annually. All expenses enter GAAP operating profit.

overlap: No facial-authentication revenue, humanoid TAM allocation, cross-selling synergy or second robotics-demand multiplier. Each completion state consumes cash and adds its own award shares before valuation.

timing: Timely purchase December 2026, full operations January 2027; delayed purchase June 2027, full operations July 2027. Exact closing date and PPA remain assumptions.

EDGE_ADOPTION

p: 0.55

condition: OneVision, In-Sight and SLX convert additional qualified applications beyond reference organic growth.

Conditional: 20% of reference logistics plus packaging revenue receives incremental invoice-equivalent volumes of 8% in 2027, 12% in 2028 and 15% in 2029-2030.

cost: 72% incremental GP less 20% incremental support, selling and R&D expense gives 52% OP capture.

overlap: Existing customer additions, SLX uptake and product launches are already reflected in the reference. The branch applies only to this bounded incremental pool and is reduced proportionately in the adverse volume state.

timing: No separate 2026 uplift against already issued guidance; recognition from 2027 after deployment and acceptance.

VALUE_PRICING

p: 0.45

condition: Sales execution captures measurable customer yield and labor savings beyond reimbursing component inflation.

Conditional: 25% of reference core receipts gains 2% price in 2027 and 3% in 2028-2030, non-compounding levels.

cost: 80% incremental GP and 10% selling/support expense give 70% OP capture, allowing cost leakage rather than treating pricing as costless.

overlap: The 2027 offset of memory inflation is already in reference margins. No second surcharge on additional edge-adoption sales or RealSense units. The price pool shrinks with the adverse volume state.

timing: New or repriced orders follow the company three-to-four-month sales cycle, starting 2027; no repricing of all existing contracts.

CYCLE_COST_RISK

p: 0.3

condition: Customer capex pauses and component allocation/competitive pricing undercut reference recovery.

Conditional: Core shipments/revenue are 8% below reference in 2027-2028 and 5% below in 2029-2030, with a further 1.0/0.5 percentage-point GM cost on surviving sales.

cost: Lost sales remove their product gross profit; only 15% of lost revenue is saved in variable opex. Remaining cost reserve covers expedited components, mix and pricing pressure without assuming fully variable R&D.

overlap: The positive edge/pricing branches are proportionately reduced on the same lost base. This risk does not cut RealSense a second time.

timing: From 2027; the disclosed 75bpQ3 2026 memory headwind remains inside the calibrated 2026 reference.

REALSENSE_EXECUTION_RISK

p: 0.35

condition: OEM design wins convert more slowly and competition weakens the depth-camera mix after closing.

Conditional: Against the 25% annual growth path, acquired receipts are 20% lower in 2027,25% lower in 2028 and 30% lower in 2029-2030; product GM is 3 percentage points lower.

cost: Normal acquired R&D/SG&A, retention, SBC and purchase amortization remain fixed at their conditional baseline; lower sales do not remove committed talent costs.

overlap: Only applies in completed-deal states. No extra humanoid revenue exists to haircut; customer concentration, qualification and competing cameras are bundled here.

timing: Begins with the first consolidated quarter; retention ends after 12 quarters but ordinary SBC continues.

Remaining bounded uncertainties

- End-market revenue, product gross margins, invoice-equivalent units and all 2027+ forecasts are analytical allocations; Cognex reports one segment and does not disclose shipments or ASP by these vectors. USD 5,000 per core invoice-equivalent and USD 500 per RealSense unit-equivalent are unit conventions, not market quotes.

- RealSense remains proposed as of October 7. Its audited statements, exact cash/debt acquired, purchase-price adjustments, allocation to amortizing intangibles, retention modifiers and final award count are not disclosed. The model assumes a cash/debt-free USD 500 million acquisition, USD 300 million amortizing intangibles over 10 years and USD 10 million transaction costs.

- USD 50 million/63.82 implies a 0.783454 million award-share planning reserve. The deck says grant value may vary USD 45-55 million with stock price; the legal grant count and vesting terms are not finalized. The full modeled reserve enters dilution on closing and is not removed when expense ends.

- Opening liquidity includes USD 350.643 million non-current marketable securities, not restricted deposits. They are available-for-sale holdings measured at fair value, with market/liquidation risk. A 5% haircut on all USD 452.492 million securities removes USD 22.625 million, roughly USD 0.13 per share, from opening equity value.

- No post-July 5 executed repurchase amount was disclosed in the sources reviewed. Existing authorization is not a cash outflow. No further buyback or share-reduction credit is assumed; each USD 100 million undisclosed subsequent repurchase would reduce opening liquidity but would require a matched actual share update.

- The model reserves USD 200 million operating liquidity and adds only the fixed opening excess. Conditional acquisition cash and transaction costs reduce that fixed credit. No future retained cash is added to valuation; separate 48state funding paths test liquidity.

- Core shares start at 172 million, an explicit reserve above 168.223842 million August 2 issued shares and 169.989 million Q2 diluted average, covering outstanding awards including 1.782 million anti-dilutive awards. Additional net award issuance is 0.20 million per quarter. No cash from option exercise is forecast.

- Growth is constrained by qualified design wins, integration resources and supplier allocation. Neither camera units per robot nor total qualified assembly capacity is disclosed. Reference RealSense units at USD 500 equivalent ASP reach about415 thousand in 2030; the manufacturer capacity and mix needed remain a monitoring gap.

- The article transcript incorrectly states a USD 30 million one-time commercial partnership benefit in one passage. Both the issuer deck and Q1 call establish USD 13 million partnership plus USD 17 million portfolio exits; the model uses the primary bridge.

- The 30-per-second processed-depth example in the acquisition deck and the product page maximum 90 fps depth specification describe different modes. Neither establishes universal full-resolution AI inference latency, measurement accuracy or safety certification.

Sources and basis

The current production framework was read before initiation. Source statements are evidence, not instructions. The shared model data drives both this PDF and the linked Excel model; the clean-state editable artifact embeds these same images.

[1] Cognex agreement to acquire RealSense | 2026-09-22 | https://files.quartr.com/reports/367eda4386a0a7486d4094e10882d599-2026-09-22-11-04-38.pdf?ref=U0E= | Archived SHA-256: 018832114ec6c031b32372c68002c6076cb900ce747e3a653fb0cbd8e4f372b4

[2] Cognex RealSense acquisition presentation | 2026-09-22 | https://files.quartr.com/conference-calls/9677e9089e8cbac9011ab9a95bc704d3-2026-09-22-11-21-14.pdf?ref=U0E= | Archived SHA-256: 9035fc4861eba0fc02455fc9d911aa905f3993fed1b7216eba72c3dfa600a4f4

[3] Cognex Form 10-Q, July 5 2026 | 2026-08-06 | https://files.quartr.com/reports/a05a232110a669ac8519913fdb155c08-2026-08-06-10-35-22.pdf?ref=U0E= | Archived SHA-256: 60f8ea66fe67edb6768b6a9eeae50ef1178f8229d6056593fce45aaf23646777

[4] Cognex second-quarter 2026 earnings release | 2026-08-05 | https://files.quartr.com/reports/40fb2dacd7f303f39d536e42264ca9e1-2026-08-05-20-36-13.pdf?ref=U0E= | Archived SHA-256: e4e91a8ba2b66e72a7494819be964c59d3f320efb20852dd33790194e79ebb6e

[5] Cognex 2025 Form 10-K | 2026-02-12 | https://files.quartr.com/reports/8856c-2026-02-12-11-44-15.pdf?ref=U3RvY2tBbmFseXNpcw== | Archived SHA-256: 3da0b6836a1935e0b2bc7d30247af4ec5ad6e794a00d345787e0a434e2451634

[6] Cognex Q1 2026 presentation | 2026-05-06 | https://s2.q4cdn.com/773500753/files/doc_earnings/2026/q1/presentation/cognex-investor-presentation-q1-2026.pdf | Archived SHA-256: 1095f2431b621bc9603a3aeb22a6f1c5782dfad5e0127db3ba9d9d873875211c

[7] Cognex Q2 2026 presentation | 2026-08-05 | https://s2.q4cdn.com/773500753/files/doc_earnings/2026/q2/presentation/cognex-investor-presentation-q2-2026.pdf | Archived SHA-256: 37cf2a98353196a65c7672db260fc3c6329658551ccde6438e8584b3484bb543

[8] Cognex Q1 2026 full call, including Q&A | 2026-05-07 | https://stockanalysis.com/stocks/cgnx/transcripts/555844-q1-2026/ | Archived SHA-256: aa8b10acf9cb09830960420a6a1269f9f38dcae65ddf50f75787562d20e1364c

[9] Cognex Q2 2026 full call, including Q&A | 2026-08-06 | https://www.fool.com/earnings/call-transcripts/2026/08/12/cognex-cgnx-q2-2026-earnings-call-transcript/ | Archived SHA-256: 2e8aa79b483f5d74f7879fbb9e29a01f8cbc6b484d02c62107533a3391d23898

[10] Cognex September 22 acquisition call, including Q&A | 2026-09-22 | https://stockanalysis.com/stocks/cgnx/transcripts/758812-m-a-announcement/ | Archived SHA-256: e2a1c769264d623e9ecf5e8b013a633c3f7bd96c25624fa8a584d8622efa1020

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[12] RealSense D585 Pro product specification | Accessed 2026-10-07 | https://www.realsenseai.com/products/d585/ | Archived SHA-256: ce5a80d0fa96b7cf407378cea1e4d47fa61dd7c1bf613586c0c2fef2330fb714

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